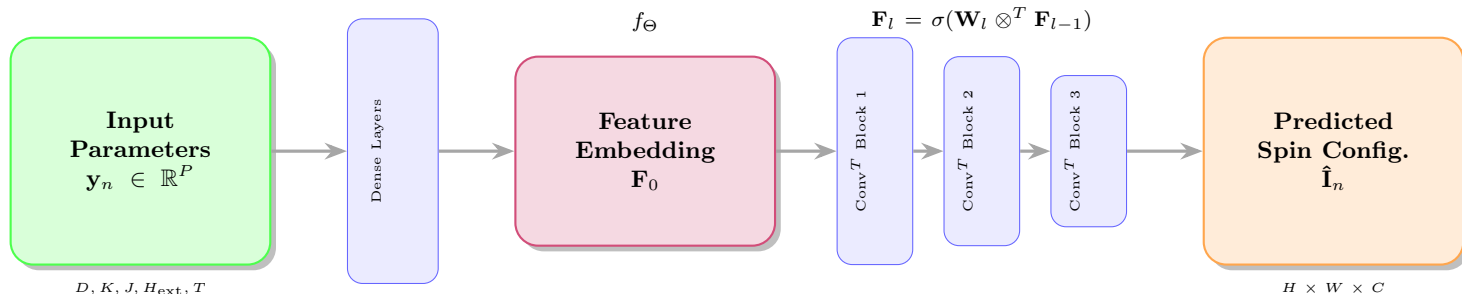


Neural Surrogate Model Architecture



Training: $\Theta^* = \arg \min_{\Theta} \mathbb{E}[\mathcal{L}(\mathbf{I}_n, \hat{\mathbf{I}}_n)]$

$$\text{Loss: } \mathcal{L}_{\text{MSE}} = \|\mathbf{I}_n - \hat{\mathbf{I}}_n\|^2$$

Strengths:

- Orders of magnitude faster
- Gradient-based optimization
- High-dim. parameter exploration

Limitations:

- Limited generalization
- Non-physical textures
- Training data dependent